Learning Goal / Big Idea / Big Question

(1) Tree-Based Learning: To employ the tree induction process to segment the feature space and construct classification/regression trees, evaluate the predictive performance of trees, and interpret trees; (2) Ensemble Learning: To explain the intuition behind ensemble learning, distinguish between the bagging and boosting methods to create ensembles, and make predictions with ensemble models.

Course Learning Objective(s)

1. Construct, evaluate, and interpret classification trees.
2. Construct, evaluate, and interpret regression trees.
3. Construct and evaluate ensembles of base learners.

Deliverables

Present your findings in the following format:

- For each question, submit a separate document (.docx or .pdf) containing a narrative of your analysis and discussion of your results.
- Submit the source files (Orange workflow and/or Python scripts) that you used to produce your results.
- Submit the files to the appropriate ICON Assignment dropbox.

Concepts to Consider

Supervised segmentation; Tree induction; Recursive partitioning; Node impurity; Entropy; Information gain; Tree pruning; Bagging; Boosting.

Assessment and Feedback Plan

Your grade depends on the accuracy and completeness of your analysis as well as the clarity of your interpretations. The level of detail should be sufficient to support the reproducibility of the results. For example, you should provide information on variables used, data pre-processing applied, algorithm parameters calibrated, as well as relevant charts and output. Feedback will be provided via SpeedGrader in the corresponding ICON Assignment dropbox.

Grading Breakdown:

Question 1: 45 points
Question 2: 25 points

DESCRIPTION

1. Telecommunications companies providing cell-phone service are interested in customer retention. In particular, identifying customers who are about to churn (cancel their service) is potentially worth millions of dollars if the company can proactively address the reason that customer is considering cancellation and retain the customer. Data on past customers (some of whom churned and some who did not) has been collected. The variables in this data *(cellphonechurn.csv*) are listed in the following table. Set aside 20% of the data as a test set and use 80% of the data for training and validation. The task is to classify a customer as a churner (someone who will soon cancel their service) or not.

Variable	Description
AccountWeeks	number of weeks customer has had active account
ContractRenewal	1 if customer recently renewed contract, 0 if not
DataPlan	1 if customer has data plan, 0 if not
DataUsage	gigabytes of monthly data usage
CustServCalls	number of calls into customer service
DayMins	average daytime minutes per month
DayCalls	average number of daytime calls
MonthlyCharge	average monthly bill
OverageFee	largest overage fee in last 12 months
RoamMins	average number of roaming minutes
Churn	"Yes" if customer cancelled service, "No" if not

   a) Via 10-folds cross-validation, evaluate pre-pruning rules to construct an individual classification tree. Report the pre-pruning settings determined in the cross-validation experiment and the corresponding value of the AUC. Compare the AUC of this pruned tree to the AUC of the full classification tree.
   b) Using the pre-pruning settings identified in part (a), construct the classification tree on all the training/validation data. Consider a customer described by: AccountWeeks = 40, ContractRenewal = 1, DataPlan = 1, DataUsage = 0, CustServCalls = 6, DayMins = 82, DayCalls = 123, MonthlyCharge = 32, OverageFee = 11, and RoamMins = 11. How does the tree classify this observation?
   c) Via 10-folds cross-validation, evaluate settings to construct a random forest. Report the settings determined in the cross-validation experiment and the corresponding value of the AUC.
   d) Via 10-folds cross-validation evaluate settings to construct a gradient boosted ensemble. Report the settings determined in the cross-validation experiment and the corresponding value of the AUC.
   e) Compare the cross-validation performance of the classification models in parts (a), (c), and (d) and select a final model. Construct this final model on all the training/validation data and report its performance measures (confusion matrix metrics, AUC, lift) on the test set.

f) Suppose the telecommunications company wishes to use the model selected in part (e) to identify customers who are likely to churn and then intervene to prevent losing these customers. Assume that the cost of the intervention (e.g., a promotional discount given to a predicted churner) is $200, that a customer always accepts the intervention and remains with the company, and retaining a customer is worth $1000. In the ROC Analysis widget for the test set, use the Analysis region (selecting Show performance line and setting FP Cost, FN Cost, and Prior probability parameters) to determine the probability threshold for churning above which a customer should be offered the promotional intervention to entice the customer to remain with the company (not churn).

g) Construct the final model selected in part (e) on all the observations in *cellphonechurn.csv*. Apply this model to the data in the file *currentcustomers.csv* to predict the churn likelihood for a set of current cell-phone customers. Based on the probability threshold determined in part (f), how many of these customers should be offered the intervention? Provide a list of the ten customers predicted most likely to churn.

2. Palanquin Auto is a national pre-owned car retailer that offers an online marketplace for customers interested in purchasing a pre-owned automobile. Palanquin collects data on automobile sales in an effort to set fair, competitive prices. The file *palanquin.csv* contains these data for the variables listed in the following table. Set aside 50% of the data as a test set and use 50% of the data for training and validation. The task is to predict a vehicle's sales price.

Variable	Description
Price	Sales price of vehicle
Year	Model year of vehicle
Manufacturer	Name of company manufacturing vehicle (categorical)
Model	Model name of vehicle (categorical)
Condition	Categorical variable describing condition of vehicle (fair, good, excellent, like new, new)
Cylinders	Number of cylinders in engine
Fuel	Type of fuel used by vehicle (diesel, electric, gas, hybrid)
Odometer	Number of miles on vehicle's odometer
Drive	Categorical variable indicating drivetrain (four wheel drive, front wheel drive, rear wheel drive)
Type	Categorical variable indicating
Paint	Color of vehicle's paint
State	State of the United States in which vehicle was sold

   a) Via 10-folds cross-validation, evaluate pruning to identify an individual regression tree. If using Orange, identify the pre-pruning settings determined in the cross-validation experiment and the corresponding value of the RMSE. If using Python, identify the cost-complexity pruning sets determined in the cross-validation experiment and the corresponding value of the RMSE. Compare the RMSE of the pruned regression tree to the RMSE of the full regression tree.
   b) Using the settings identified in part (a), construct the regression tree on all the training/validation data. Using this regression tree, predict the sales price of a vehicle with the following variable values.

Year	2016
Manufacturer	Dodge
Model	Charger
Condition	like new
Cylinders	6
Fuel	gas
Odometer	62,000
Drive	rwd
Type	sedan
Paint	silver
State	Alabama

c) Via 10-folds cross-validation, evaluate settings to construct a random forest. Report the settings determined in the cross-validation experiment and the corresponding value of the RMSE.
d) Via 10-folds cross-validation, evaluate settings to construct a gradient boosted ensemble. Report the settings determined in the cross-validation experiment and the corresponding value of the RMSE.
e) Compare the performance of the regression models in parts (a), (c), and (d) in the cross-validation experiment and select a final model. Construct this final model on all the training/validation data and determine its value of the RMSE on test set. How does the test set RMSE compare to the RMSE from the corresponding cross-validation experiment?